The Role of Networks in Semantic and Morphological Processing

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Department of Psychology, National University of Singapore

Dr. Cynthia Siew

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Introduction

The application of psychology to language, known as the field of psycholinguistics, theorizes the existence of a mental lexicon that is accessed when we use language. The mental lexicon comprises of mental representations of words which hold their internal properties, as well as their external relations with other words, on features such as visual form (orthography), sound (phonology), meaning (semantics) and structure (morphology). This mental lexicon is often conceptualized as a 'computational' network; however, theorists and models differ on the exact hierarchical structure and rule-based processes that occur within the lexicon (Coltheart, 2006; Forster & Taft, 1976; Mehler et al., 2015; Seidenberg, 2005; Taft & Forster, 1975; Vitevitch, 2008).

In the following paper, I aim to give a brief overview of the major models of psycholinguistic networks as well as their applications in the subfields of semantic and morphological analysis. Further, I will consider the main body of evidence for each model, and discuss some gaps in the literature that must be addressed to better understand the cognitive processes underlying the use of language in the mental lexicon.

Models

The networks-based models of psycholinguistics rose in tandem with the rise of the information-processing approach in psychology as a whole. According to the information-processing approach, the human mind acts as a sort of computer that "transformed, reduced, elaborated, stored, recovered, and used" sensory input (Neisser, 1967, p. 4). From this approach stemmed the parallel distribution processing (PDP) or connectionist model of language

processing, wherein linguistic knowledge, rather than being a static set of internalized information and rules, becomes an interconnected network of dynamic nodes that respond to learning from the environment thorugh feedback mechanisms. These connectionist models became the basis of psycholinguistic networks as we know them today. In the traditional generalist connectionist model, language structure involves layers of nodes organised by groups of properties found in word processing (semantics, orthography and phonology); further, these groups are connected to each other hierarchically, where aspects such as orthography and phonology are 'low-level' are hence more easily accessed than semantic information, which entails a longer processing path (Seidenberg, 2005; see Figure 1). Early evidence for this network comes from an algorithm which was configured to the assumptions of the connectionist model. This algorithm was able to simulate the connectionist model of human learning in word-recognition – the model essentially 'taught' itself to distinguish between word and nonwords over time and exposure to data (Harm & Seidenberg, 1999, as in Seidenberg, 2005).

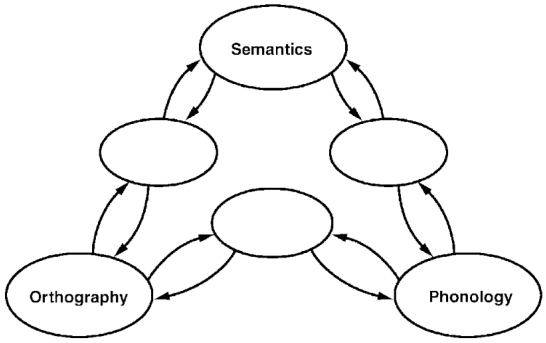


Fig. 1. Theoretical framework (introduced by Seidenberg and McClelland, 1989) that has served as the basis for several implemented models of word reading. Collectively known as the ''triangle model,'' the models differ in details and focus but are based on the same theoretical principles. Large ovals represent groups (''layers'') of units that encode different types of information: orthography (spelling), phonology (derived from pronunciation and sound), and semantics (meaning). Smaller ovals represent ''hidden units,'' which increase the computational capacity of the network and provide the basis for abstraction. Most models have focused on the orthography-to-phonology mapping. Harm and Seidenberg (2004) implemented both orthography–semantics and orthography–phonology–semantics components, using a variant of this architecture.

Figure 1. The connectionist model for word processing, as seen in Seidenberg, 2005.

However, as research developed in the field, certain effects which were observed in human behavioural data fit less and less with the current models. There were differential effects observed within the micro-networks of semantics, orthography and phonology themselves; moreover, the proposed networks that accommodated these behavioural effects more often than not could only explain effects in their semantic, orthographic or phonological domain and were not able to apply themselves to all linguistic processing. As rightly pointed out by Mahler at al. (2015, p. VI), this has led to the current state of networks analysis which provides a range of "laws of information processing without clarifying their synergetic interdependencies".

The present review focuses on the domains of semantics and morphology in linguistic processing as they remain the most contentious fields with regards to both behavioural data and subsequent network formulations. In the case of semantics, the 'higher' level of representation it holds in networks have been called into question, particularly in regards to semantic associations between words separated by multiple nodes, and the directions of these associations. For morphology, the controversy centres around the difficulty in extracting the exact role of morphology independent from the related properties of orthography, phonology and semantics, which all contribute to morphological relatedness.

Semantic Networks. The most popular model of semantic networks is the spreading activation model (Collins & Loftus, 1975). Here, words represent a node in the mental lexical network and edges represent a shared meaning or semantic association between them (see Figure 2). The model provides for two main rules: first, activation of a single node sends activation to other, connected nodes and second, this activation is a limited resource which decreases with

distance across the network, and hence benefits the closest words the most. The activation of ‘red’ activates ‘fire’ (among other nodes), which may in turn weakly activate ‘house’. However, ‘fire engine’ benefits from a stronger activation because it may receive activation directly from ‘red’ and ‘fire’. Little, if any, activation will be available for peripheral nodes like ‘street’ and ‘clouds’.

The controversy regarding this model involves the degree to which activation spreads beyond a single node. It has been well-established that sharing a node can facilitate some activation in priming tasks – that is, when ‘fire’ is presented for processing, the subsequent processing of ‘cherries’ is faster, presumably due to the activation generated from the node ‘red’ in between them. However, this effect has not been replicated for words that are more than two steps away from a prime – that is, ‘fire’ does not benefit the processing of ‘clouds’ in any way, equivalent to a completely unrelated word.

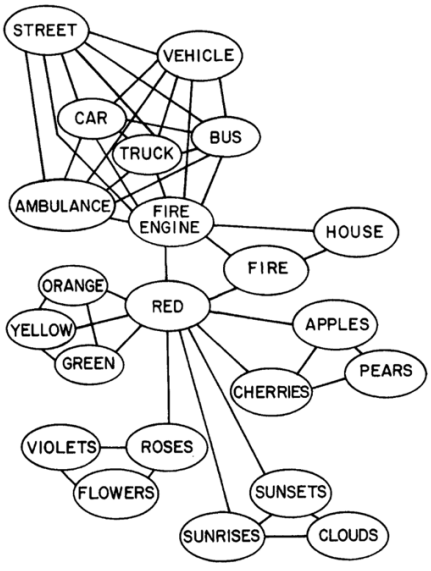


Figure 2. The semantic association network as in Collins and Loftus (1975).

Network analysis of semantic associations typically link nodes if at least two participants have made such an association response to the target word (Steyvers & Tenenbaum, 2005). However, no differences of semantic similarity or distance are reflected in these networks – whether ‘fire’ is very strongly related to ‘red’ as compared to ‘roses’ is not taken into account, only whether they are connected or not. The need for weighted association networks has since been made evident (Christensen & Kenett, 2021; Kenett et al., 2011, 2017); however these weighted networks in turn are computed in terms of correlations, making the analysis necessarily undirected. It is difficult to represent both a strength and direction of association in semantic priming networks as was posited by Collins and Loftus (1975), making it difficult to test the theoretical assumptions put forth by them. Further evidence from behavioral data also shows that direction is important in semantic priming: forward associations and backward associations between nodes have been shown to significantly affect the automatic semantic processing of a word (Heyman et al., 2015). However, the analysis of separate directed networks and weighted networks in a regression model have been able to account for some variance in behavioral data regarding semantic priming tasks (Kumar et al., 2020).

Morphological Networks. A morpheme is a word that has meaning of itself and cannot be further broken down into any other meaningful constituents, such as ‘depart’. Morphemes can be added to each other to create words, such as the addition of ‘ed’ to make ‘departed’ or ‘ure’ to make ‘departure’. It can be said that these words are simply different forms of each other. However, the analysis of morphological processes is complex due to the fact that morphologically related words are also orthographically, phonologically and semantically related. For example, ‘depart’ shares letters, sound and meaning with its form ‘departure’. Researchers attempt to study morphology by comparing words that don’t share meaning with but

are still morphologically related to their roots, such as ‘department’ and ‘departure’ which are morphologically but not semantically related (see Figure 3).

Behavioral research in the field of morphology is divided on whether words such as ‘department’ and ‘departure’ are equally primed by ‘depart’ – some studies say yes (Andrews & Lo, 2013; W. D. Marslen-Wilson et al., 2008; Morris et al., 2007; Kathleen Rastle et al., 2004) while other studies say no (Feldman et al., 2004; Lavric et al., 2007; Rastle et al., 2000, 2004; Whiting et al., 2017). The application of networks analysis to these results have not resulted in any conclusive evidence – the literature suggests that it is likely that the interdependence of morphology and semantics may be a nonlinear effect (see Feldman & Milin, 2019). However, some studies, notably that of Milin et al. (2017), have been able to provide support for hybrid models that attempt to find a mediating effect of semantics on morphological priming (Baayen et al., 2011).

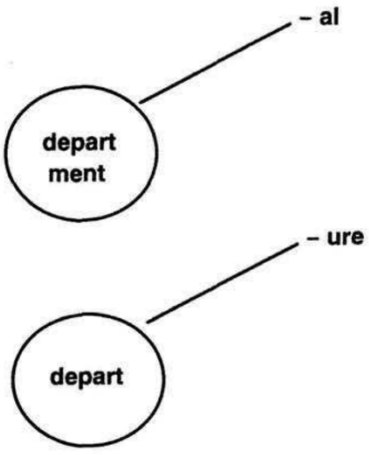


Figure 6. Lexical entries for a cognitively monomorphemic word (*department*), shown here in combination with the affix *-al*, and for a morphologically complex form (*departure*).

Figure 3. The morphological relatedness network as in Marslen-Wilson et al., 1994.

Limitations and Future Directions

All network constructions are limited in the information they can hold. In semantic networks, the difficulty arises in computing and representing the strength and direction of associations between words in such a way that yields itself to concise network analyses. In morphology, the difficulty revolves around representing different types of relations (semantic, orthographic, phonological, morphological) words have among each other, along with the complications of different values that may be assigned to each relationship. Networks have predictive power when the scope of effects being tested is narrow – such as explaining a semantic or morphological priming effect model observed for certain sets of words. However, computing a network that can hold different types of relations (semantic and morphological), each of which have different directions and associated values, is not easy. It is thus useful to apply networks in order to fit specific data to a model, but difficult to rely on network analyses to accurately represent the mental lexicon as a whole.

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